SUPPLEMENTARY MATERIALS

Supplementary Results

1. Details of model training and experimental setup

Our model was trained for 200 epochs with an initial learning rate of 5e-4. We implemented a learning rate scheduler to reduce the rate by a factor of 0.95 every five epochs, promoting optimal convergence and fitting. The Adam optimizer was employed for efficient fitting, complemented by an early stopping strategy to mitigate overfitting by ceasing training when no accuracy improvements were observed after a specified number of epochs. To further prevent overfitting, dropout was incorporated, randomly excluding neurons during training to avoid dependency on particular features. Training was performed in data-parallel mode on two NVIDIA RTX 4080 GPUs, achieving an average epoch time of 2.1 s and a peak GPU memory footprint of approximately 8.5 GB per GPU.

In the final configuration, the Graph Transformer comprised 2 layers and 8 attention heads, with 128 hidden channels and 64 output channels. Both LayerNorm and BatchNorm were enabled to enhance training stability. The contrastive learning component utilized a projection head with a dimension of 128 and a temperature parameter (τ) set to 0.2. The total contrastive loss was formulated as a weighted combination of sequence-structure and label-level components, with respective weights of 0.6 and 0.4. All experiments were conducted using a batch size of 32.

For structure and sequence representations, we used the ESMFold v1 model to predict 3D structures and the ESM-2 model (esm2_t12_35M_UR50D) to extract sequence embeddings[1]. These versions were selected to balance predictive performance and computational efficiency within our pipeline. The average prediction time of ESMFold was approximately 1.3 seconds per sequence.

Our experimental development was based on Python[2], and we used the PyTorch framework to construct and train the model[3].

[1] Lin, Zeming, et al. "Evolutionary-scale prediction of atomic-level protein structure with a language model." Science379.6637 (2023): 1123-1130.
[2] Van Rossum, Guido, and Fred L. Drake Jr. Python tutorial. Vol. 620. Amsterdam, The Netherlands: Centrum voor Wiskunde en Informatica, 1995.
[3] Paszke, Adam, et al. "Automatic differentiation in pytorch." (2017).

2. Overview of amino acid distributions

To investigate the sequence length distribution of our collected data, as well as the differences in amino acid composition between CTPs and non-CTPs, and among different functional types of CTPs, we conducted a detailed analysis.

Figure S1(A) shows the sequence length distribution of positive samples (CTPs) and negative samples (non-CTPs). The results indicate that the majority of CTP sequences range between 10 and 30 amino acids. Moreover, the sequence length distribution between CTPs and non-CTPs is relatively consistent, suggesting that the model does not rely on sequence length as a primary decision-making feature. Similar patterns were observed under the CD-HIT 0.9 and 0.8 thresholds, as shown in Figure S2(A) and Figure S3(A).

Figure S1(B) compares the average amino acid composition (AAC) of CTPs and non-CTPs, grouped by the physicochemical properties of the amino acids. The results reveal that CTPs are enriched in certain hydrophobic amino acids (e.g., Leucine (L)) and basic amino acids (e.g., Lysine (K) and Arginine (R)). In contrast, non-CTPs show higher levels of polar acidic amino acids such as Aspartic acid (D) and Glutamic acid (E), indicating distinct physicochemical properties between these two groups. Comparable trends were also observed in Figure S2(B) and Figure S3(B), supporting the consistency of these findings across different redundancy levels.

Figure S1(C) further explores the AAC of six functional subtypes of CTPs, including Tumor active, Cancer-targeted, Membrane-targeted, Cell-penetrating, Membrane lysis, and Induce apoptosis peptides. Each functional type exhibits unique amino acid distribution patterns. For example, Membrane lysis peptides are enriched in hydrophobic amino acids like Leucine (L) compared to other types, whereas Cancer-targeted and Cell-penetrating peptides show higher levels of basic amino acids such as Arginine (R). These findings suggest that each functional subtype of CTPs may have specific sequence preferences, reflecting their distinct biological roles and mechanisms of action.

3. Ablation experiments

3.1. Comparison of different node features in the protein graph

To identify the optimal combination of node features in the protein graph, we evaluated several feature combinations. Firstly, we compared the influence of different physicochemical property (PCP) combinations on the model's performance. The properties evaluated included Zscale, Hydrophobicity, Isoelectric point, and Molecular weight. As shown in Table S5, among the individual features, Hydrophobicity achieved the highest accuracy and MCC, reaching 0.902 and 0.804, respectively. Molecular weight obtained the best sensitivity and AUC, with values of 0.936 and 0.948, respectively. By combining all four features, the model achieved improved overall performance, with an accuracy of 0.911 and an MCC of 0.822, offering a more balanced performance across the metrics.

Next, to further enhance the model's performance, we incorporated evolutionary information into the PCP features. The evolutionary information was drawn from both BLOSUM and PAM matrices, using various versions: BLOSUM45, BLOSUM62, BLOSUM80, PAM30, PAM70, and

PAM250. The inclusion of evolutionary information consistently improved the model's overall performance. Notably, when using BLOSUM62, the model achieved the best performance, with an accuracy of 0.926 and an MCC of 0.852.

Overall, the integration of physicochemical properties with evolutionary information significantly enhanced the model's performance. These results suggest that combining multiple feature types is crucial for improving the predictive power of the model.

3.2. Ablation analysis of model architecture components

We conducted a comprehensive series of ablation experiments to systematically assess the contributions of each core component within the GeoCTP framework. Specifically, we evaluated seven model variants: (1) without the sequence features (w/o Sequence Features), (2) without the structural encoder (w/o Graph Transformer), (3) without the graph attention mechanism (w/o Graph Attention), (4) without the sequence–structure contrastive learning (w/o Seq-Struct Contrast), (5) without the label-level contrastive learning (w/o Label Contrast), (6) without the overall contrastive learning module (w/o Contrastive Learning), and (7) the full model (GeoCTP (Full model)). All variants were trained and evaluated using the same dataset and protocol for a fair comparison.

As shown in Table S6, the full model consistently achieved the best performance across all metrics (accuracy = 0.926, MCC = 0.852, AUC = 0.958), highlighting the synergistic effect of all components. Removing the contrastive learning module led to a noticeable performance drop (MCC = 0.831, AUC = 0.950), confirming the importance of both modality-level and label-level alignment in enhancing feature discrimination. Excluding the graph attention mechanism also degraded performance (MCC = 0.823), underscoring the role of multi-head attention in modeling complex inter-residue interactions within structural graphs. Notably, removing the sequence feature branch resulted in a significant decrease in accuracy (0.883) and MCC (0.773), demonstrating the critical importance of semantic features derived from pretrained protein language models. On the other hand, excluding the Graph Transformer led to a marked drop in specificity (0.895) and MCC (0.813), indicating that structure-aware modeling via PDB-derived graphs substantially enhances the model's ability to discriminate negative samples.

3.3. Comparison between single-task and multi-task learning framework

To assess whether a multitask learning framework could improve model performance, we implemented an architecture with a shared encoder and task-specific heads for simultaneous binary and multi-label classification. We then compared this multi-task model with the original single-task approach, in which each task was trained independently.

As shown in Table S7 and Table S8, the single-task model consistently outperformed the multitask variant across both tasks. For binary classification, the single-task model achieved higher accuracy (0.926 vs. 0.907), MCC (0.852 vs. 0.815), and AUC (0.958 vs. 0.950). Similarly, in the multi-label task, it showed better Aiming (0.905 vs. 0.870), $\text{accuracy}_{\text{MLC}}$ (0.829 vs. 0.801), and lower Absolute False rate (0.075 vs. 0.091).

These results suggest that joint training introduces task interference due to gradient conflicts and differing optimization dynamics between the binary and multi-label objectives. Moreover, the limited encoder capacity may not sufficiently capture the task-specific nuances, especially given the differing biological focus of each task. Overall, the single-task setup provides a more stable and effective solution for CTP prediction in our framework.

3.4. Evaluation of different negative sampling strategies

We conducted a comparative analysis of three negative sampling strategies: (1) random sampling, (2) length-composition matched sampling (similarity-based), and (3) hard negative mining based on model confidence scores. As summarized in Table S9, GeoCTP exhibited consistent performance across these strategies, indicating robustness to variations in negative sample construction.

Among them, we adopted the length-composition matched strategy, as it effectively reduces distributional bias between CTPs and non-CTPs. Without such control, models may overfit to superficial differences, such as sequence length or amino acid composition, rather than learning functionally relevant features.

We excluded random sampling due to its susceptibility to uncontrolled bias. Hard negative mining, though capable of increasing learning difficulty, risks introducing peptides with unrecognized anticancer potential, leading to label noise. It is also more computationally expensive and less reproducible. In contrast, our chosen strategy offers a balanced trade-off between biological relevance, statistical consistency, and practical feasibility.

4. Comparison of baseline machine learning and deep learning models

To evaluate the performance of GeoCTP, we compared it with a series of classical machine learning and deep learning baseline models. For the machine learning baselines, we utilized a combination of handcrafted features widely used in peptide classification tasks, including Amino Acid Composition, Dipeptide Composition, Pseudo-Amino Acid Composition, Composition of k-Spaced Amino Acid Pairs, and Composition of k-Spaced Amino Acid Group Pairs. The classifiers employed included Random Forest, Extra Trees, Gradient Boosting, AdaBoost, and K-Nearest Neighbors.

For the deep learning baselines, protein sequences were encoded using one-hot representations and input into standard architectures, including Convolutional Neural Networks (CNN), Multi-Layer Perceptrons (MLP), Bidirectional Long Short-Term Memory (BiLSTM), Transformer, and graph-based models such as Graph Convolutional Networks (GCN) and Graph Attention Networks (GAT).

As illustrated in Figure S6 and Tables S14-15, GeoCTP significantly outperformed all compared methods across multiple evaluation metrics. Among machine learning models (Figure S6A), Gradient Boosting achieved the highest MCC of 0.772 and AUC of 0.942, whereas the other models showed relatively lower and more variable performance. Among deep learning models (Figure S6B), Transformer achieved the strongest baseline results with an accuracy of 0.879,

MCC of 0.761, and AUC of 0.932, while BiLSTM exhibited significantly lower performance. GeoCTP achieved the best overall metrics with an accuracy of 0.926, MCC of 0.852, and AUC of 0.958, demonstrating its strong capability to integrate sequence and structure information and outperforming both conventional and state-of-the-art models.

5. Visualization of embedding alignment with contrastive learning

To evaluate the effectiveness of contrastive learning in aligning sequence and structure embeddings, we conducted a visual analysis using t-distributed Stochastic Neighbor Embedding (t-SNE). We projected the high-dimensional representations of both modalities into a 2D space and compared their distributions under two settings: with and without contrastive learning.

As shown in Figure S7(A), in the absence of contrastive learning, sequence and structure embeddings of the same peptides are scattered and poorly aligned, with significant overlap between CTP and non-CTP samples. In contrast, Figure S7(B) illustrates that contrastive learning leads to more compact clusters and improved modality consistency. Embeddings from the same peptide across modalities are drawn closer, and clearer separation is observed between positive and negative classes.

These results indicate that our contrastive learning strategy effectively enhances the alignment between heterogeneous feature spaces, supporting its role in improving model discrimination and representation quality.

Supplementary Figures

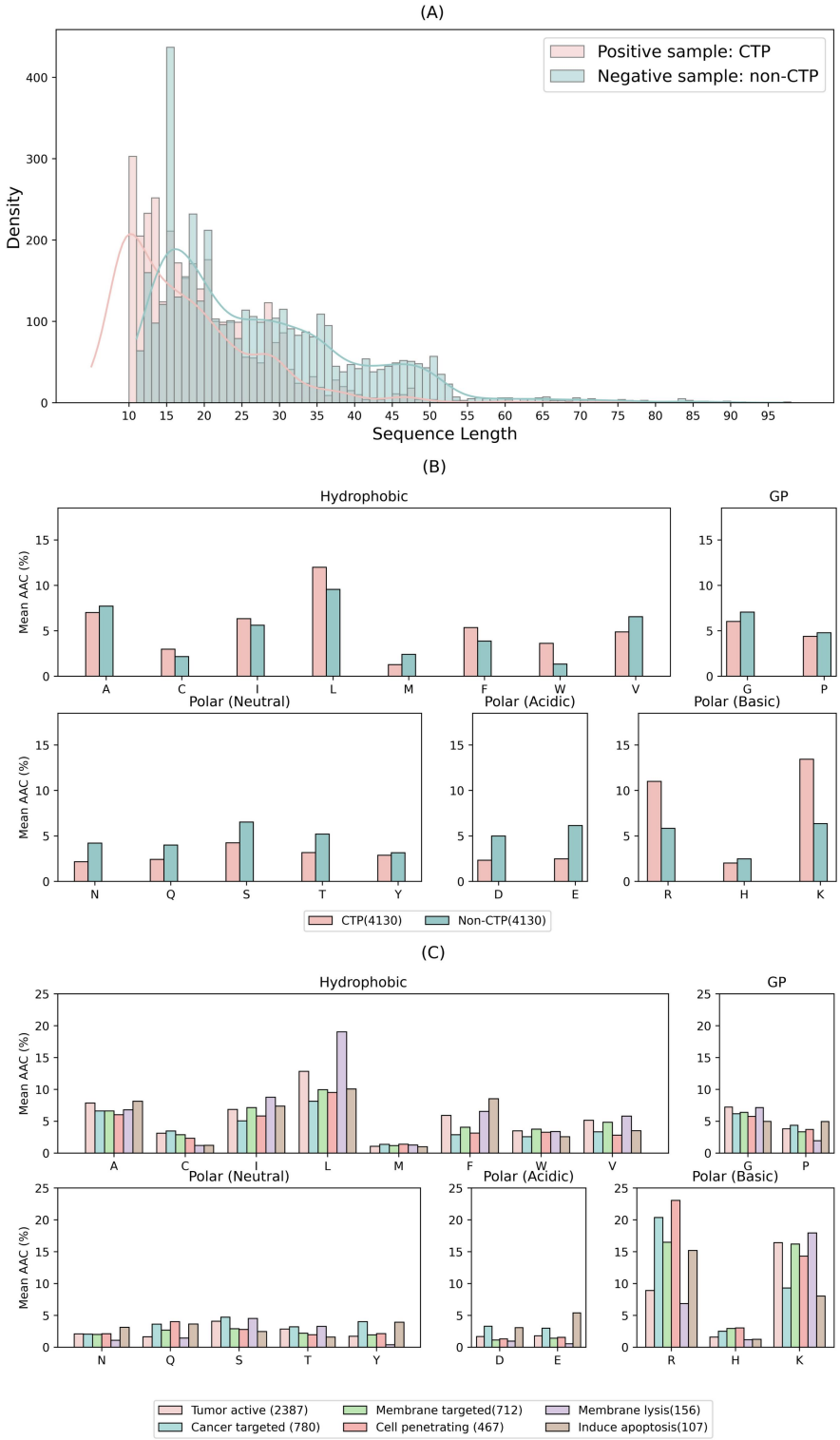

Figure S1. Statistics on the CTP dataset under the CD-HIT threshold of 1.0. (A) Length distribution of positive samples (CTP) and negative samples (non-CTP). (B) Mean AAC of positive samples (CTP) and negative samples (non-CTP). (C) Mean AAC for the six functional types of CTPs, including Tumor active, Cancer-targeted, Membrane-targeted, Cell-penetrating, Membrane lysis, and Induce apoptosis.

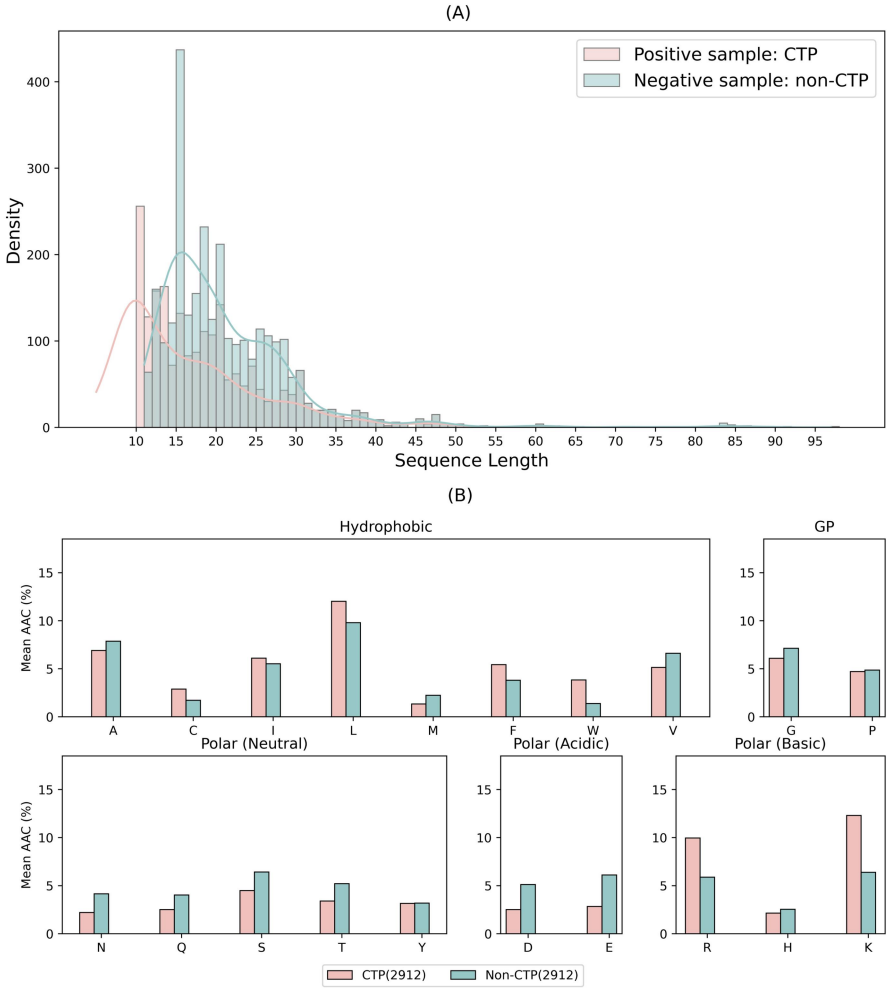

Figure S2. Statistics on the CTP dataset under the CD-HIT threshold of 0.9. (A) Length distribution of positive samples (CTPs) and negative samples (non-CTPs). (B) Average amino acid composition (AAC) profiles of CTPs and non-CTPs.

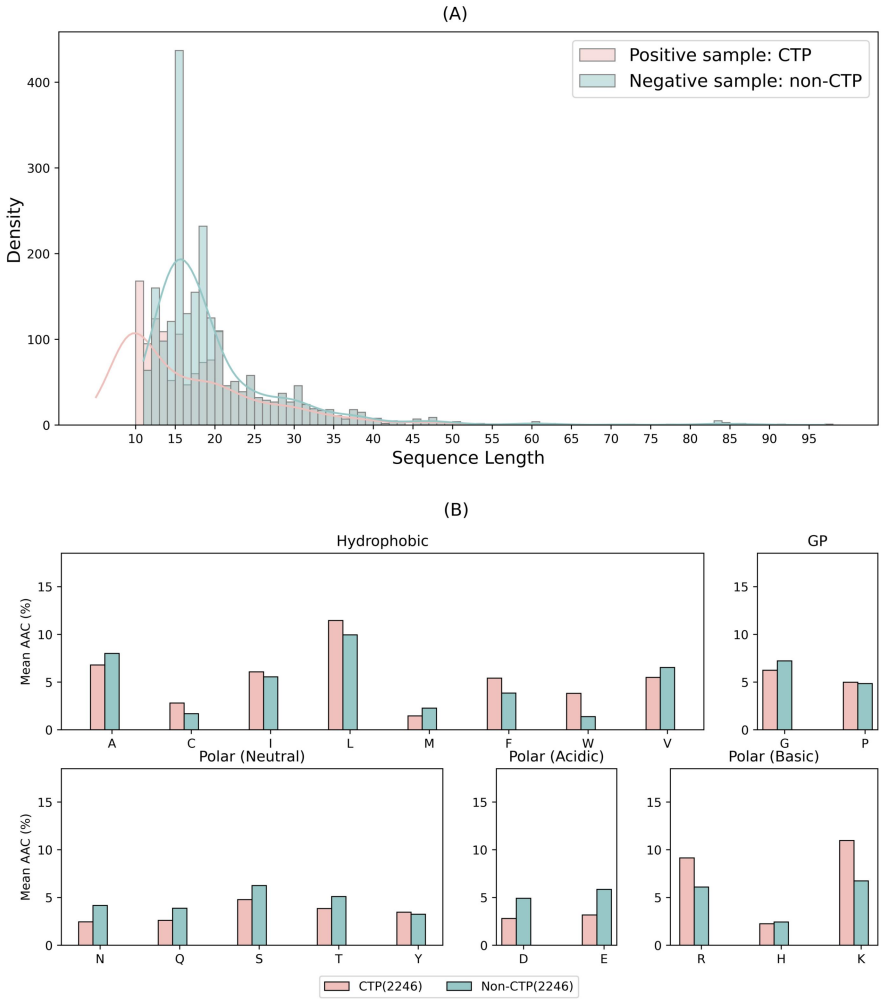

Figure S3. Statistics on the CTP dataset under the CD-HIT threshold of 0.8. (A) Length distribution of positive samples (CTPs) and negative samples (non-CTPs). (B) Average amino acid composition (AAC) profiles of CTPs and non-CTPs.

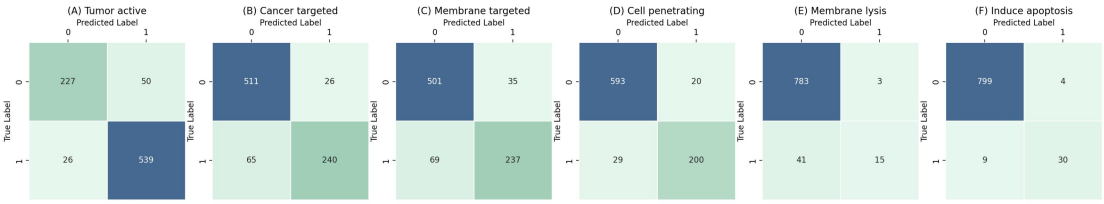

Figure S4. Performance evaluation of the model in the second stage for single-label classification tasks, including: (A) Tumor-active confusion matrix, (B) Cancer-targeted confusion matrix, (C) Membrane-targeted confusion matrix, (D) Cell-penetrating confusion matrix, (E) Membrane lysis confusion matrix, and (F) Induce apoptosis confusion matrix.

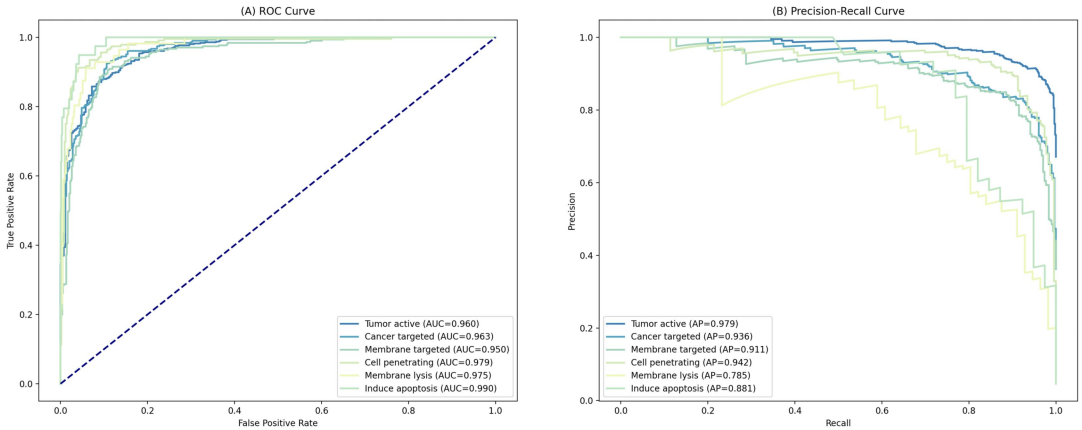

Figure S5. Performance evaluation of the model in the second stage for single-label classification tasks, including: (A) ROC curves and (B) Precision-Recall (PR) curves for each functional type.

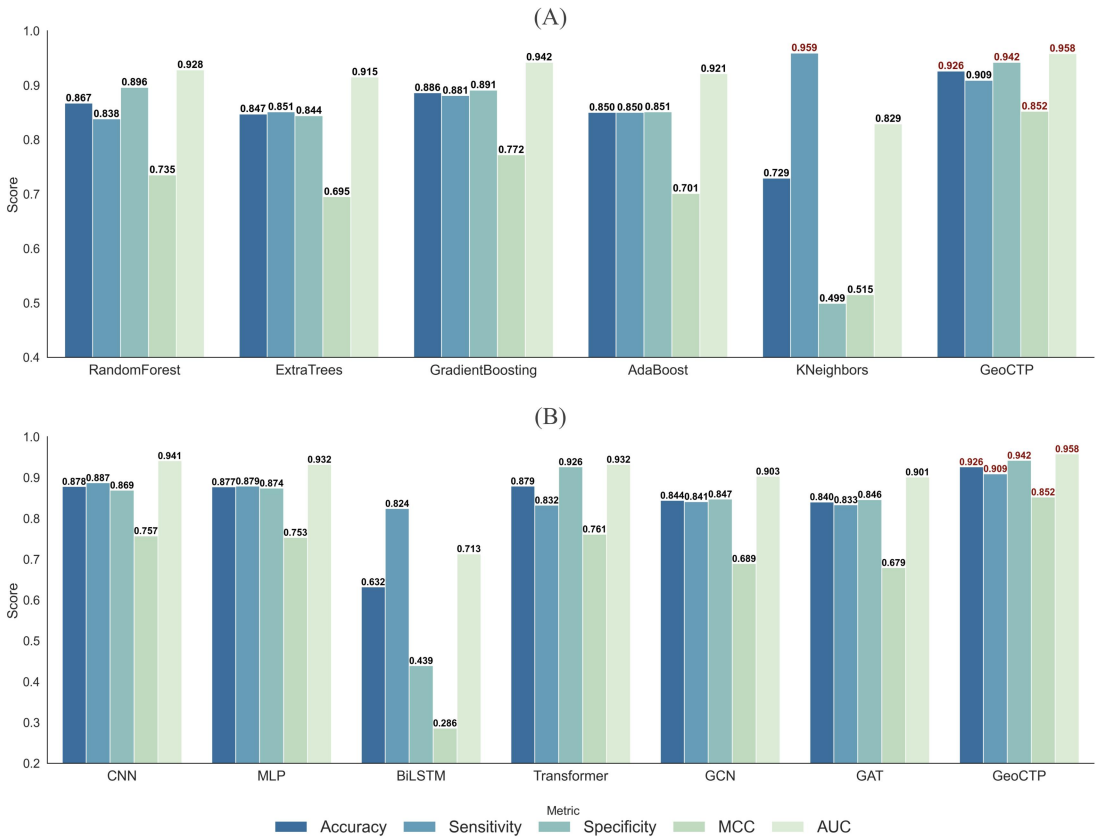

Figure S6. Performance comparison of GeoCTP with baseline models on the CTP dataset under the CD-HIT threshold of 1.0. (A) Comparison with machine learning baseline models, including RandomForest, ExtraTrees, GradientBoosting, AdaBoost, and KNeighbors. (B) Comparison with deep learning baseline models, including CNN, MLP, BiLSTM, Transformer, GCN, and GAT. Five evaluation metrics are reported: accuracy, sensitivity, specificity, MCC, and AUC. The best-performing scores for each metric are highlighted in red.

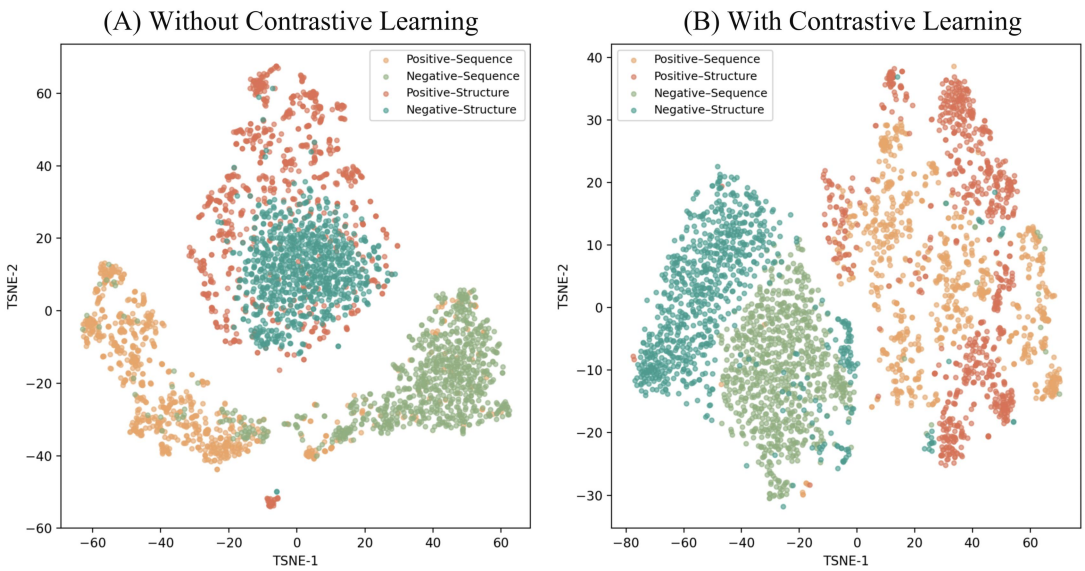

Figure S7. t-SNE visualization of sequence and structure embeddings. Embedding distributions are shown for models trained (A) without and (B) with contrastive learning.

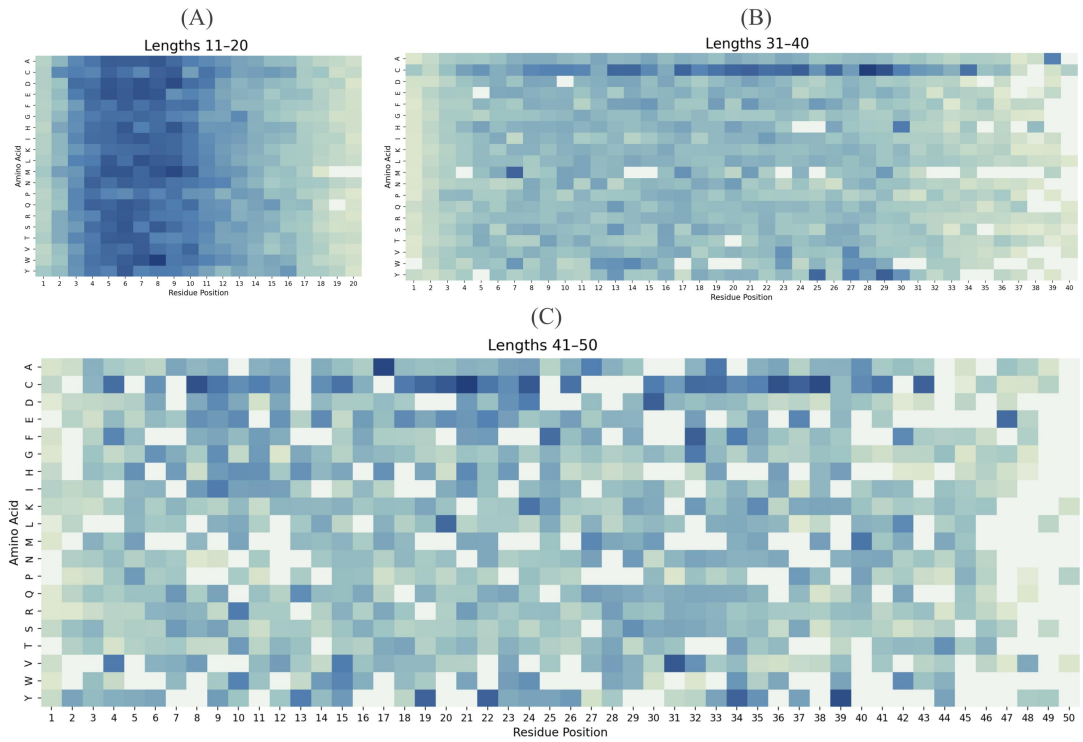

Figure S8. Attention score heat maps across different peptide length ranges. Each panel shows average attention scores aggregated by amino acid type (y-axis) and residue position (x-axis) for peptides of specific lengths: (A) lengths 11-20, (B) lengths 31-40, and (C) lengths 41-50.

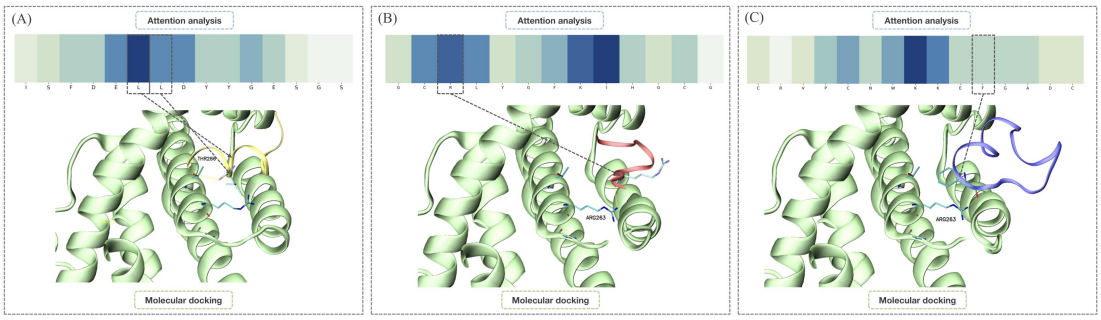

Figure S9. Three case studies of model attention and peptide-protein docking. (A-B) Correctly predicted examples in which high-attention residues participate in hydrogen-bond interactions with the target protein. (C) A high-confidence misprediction example, illustrating that attention hotspots do not always correspond to the true interacting residue.

Supplementary Tables

Table S1. Keywords for negative set filtering.

Module	Keywords	References
Primary cancer context	cancer; anticancer; anti-cancer; antitumor; anti-tumor; neoplasm; malignancy; tumor; tumour; carcinoma; sarcoma; leukemia; lymphoma; myeloma; melanoma; glioblastoma; metastasis; metastatic	1. Hanahan, Douglas, and Robert A. Weinberg. "Hallmarks of cancer: the next generation." cell 144.5 (2011): 646-674. 2. Hanahan, Douglas. "Hallmarks of cancer: new dimensions." Cancer discovery 12.1 (2022): 31-46.
Tumor-homing/targeting	tumor homing; tumor-homing; tumor targeting; tumor-targeting; cancer targeting; neoplasm targeting; tumor penetration; tumor-penetrating; tumor tropism; tumor-tropic	3. Sugahara, Kazuki N., et al. "Coadministration of a tumor-penetrating peptide enhances the efficacy of cancer drugs." science 328.5981 (2010): 1031-1035. 4. Teesalu, Tambet, Kazuki N. Sugahara, and Erkki Ruoslahti. "Tumor-penetrating peptides." Frontiers in oncology 3 (2013): 216.
Tumor microenvironment	tumor microenvironment; TME; tumor stroma; tumor vasculature; angiogenesis in tumor	5. Quail, Daniela F., and Johanna A. Joyce. "Microenvironmental regulation of tumor progression and metastasis." Nature medicine 19.11 (2013): 1423-1437. 6. Hanahan, Douglas, and Lisa M. Coussens. "Accessories to the crime: functions of cells recruited to the tumor microenvironment." Cancer cell 21.3 (2012): 309-322.
In vivo tumor models	xenograft; patient-derived xenograft; PDX; orthotopic model; syngeneic model; intratumoral injection; tumor growth	7. Hidalgo, Manuel, et al. "Patient-derived xenograft models: an emerging platform

	inhibition; tumor volume; bioluminescence imaging	for translational cancer research." Cancer discovery 4.9 (2014): 998-1013. 8. Workman, P., et al. "Guidelines for the welfare and use of animals in cancer research." British journal of cancer 102.11 (2010): 1555-1577.
Tumor-homing motifs/named peptides	RGD; iRGD; NGR; LyP-1; GE11;	9. Ruoslahti, Erkki. "Tumor penetrating peptides for improved drug delivery." Advanced drug delivery reviews 110 (2017): 3-12.
Membrane-targeted terms	membrane; membrane targeting; membrane-disruptive; membrane permeabilization; membranolytic; pore-forming; carpet model; barrel-stave; membrane rupture; membrane destabilization; membrane lysis	10. Hoskin, David W., and Ayyalusamy Ramamoorthy. "Studies on anticancer activities of antimicrobial peptides." Biochimica et Biophysica Acta (BBA)-Biomembranes 1778.2 (2008): 357-375.
Cell-penetrating terms	CPP; cell-penetrating peptide; cell penetration; internalization; cellular uptake; TAT; penetratin; transportan; oligoarginine; poly-arginine; R8; R9; Arg-rich	11. Torchilin, Vladimir P. "Tat peptide-mediated intracellular delivery of pharmaceutical nanocarriers." Advanced drug delivery reviews 60.4-5 (2008): 548-558. 12. Guidotti, Giulia, Liliana Brambilla, and Daniela Rossi. "Cell-penetrating peptides: from basic research to clinics." Trends in pharmacological sciences 38.4 (2017): 406-424.
Apoptosis-related terms	apoptosis; apoptotic; pro-apoptotic; caspase-3; caspase-8; caspase-9; mitochondrial depolarization; cytochrome c; Bax; Bcl-2; Annexin V; TUNEL	13. Elmore, Susan. "Apoptosis: a review of programmed cell death." Toxicologic pathology 35.4 (2007): 495-516. 14. Gavrieli, Yael, Yoav Sherman, and Shmuel A. Ben-Sasson. "Identification of

		programmed cell death in situ via specific labeling of nuclear DNA fragmentation." The Journal of cell biology 119.3 (1992): 493-501.
Other cancer cell-death/stress terms	ferroptosis; pyroptosis; necroptosis; ROS; DNA damage; cell cycle arrest	15. Dixon, Scott J., et al. "Ferroptosis: an iron-dependent form of nonapoptotic cell death." cell 149.5 (2012): 1060-1072. 16. Vandenabeele, Peter, et al. "Molecular mechanisms of necroptosis: an ordered cellular explosion." Nature reviews Molecular cell biology 11.10 (2010): 700-714.
Receptors/markers for tumor targeting	EGFR; HER2; ERBB2; PD-1; PD-L1; PSMA; CXCR4; TFRC; NRP1; GPC3; MET; CD44	17. Yarden, Yosef, and Mark X. Sliwkowski. "Untangling the ErbB signalling network." Nature reviews Molecular cell biology 2.2 (2001): 127-137. 18. Desgrosellier, Jay S., and David A. Cheresh. "Integrins in cancer: biological implications and therapeutic opportunities." Nature Reviews Cancer 10.1 (2010): 9-22.

Table S2. Per-model false-positive rates (FPR) of four ACP predictors on a random sample (n=200) drawn from the final negative pool.

Model	Positives / n	FPR (%)	95% CI
ACP-MHCNN	5 / 200	2.5	1.1 – 5.7
AMPfun	4 / 200	2.0	0.8 – 5.0
AntiCP3.0	5 / 200	2.5	1.1 – 5.7
ACP-CapsPred	6 / 200	3.0	1.4 – 6.4

Table S3. Ensemble FPRs under union (⩾1), soft majority (⩾2/4), and strict majority (⩾3/4) on the same sample (n=200). The union rate is interpreted as a conservative upper bound on residual contamination, whereas majority-vote rates provide a more realistic high-confidence estimate.

Rule	Positives / n	FPR (%)	95% CI
Any (union ⩾1)	10 / 200	5.0	2.7 – 9.0
Soft majority (⩾2/4 models)	6 / 200	3.0	1.4 – 6.4
Strict majority (⩾3/4 models)	3 / 200	1.5	0.5 – 4.3

Table S4. Structural overlap between negatives and positives. Overlap is defined as TM-score > 0.5.

CD-HIT Threshold (CTP)	TM min	TM median	TM mean	Proportion (TM>0.5) with 95% CI
1.0	0.118	0.313	0.389	26.22% (24.84% - 27.55%)
0.9	0.126	0.316	0.400	28.19% (26.68% - 29.84%)
0.8	0.148	0.331	0.412	28.63% (26.89% - 30.54%)

Table S5. Performance comparison of different node-level features and their combinations. The best performance in each metric is highlighted in bold.

Node feature	Accuracy	Sensitivity	Specificity	MCC	AUC
Zscale	0.899	0.919	0.879	0.799	0.942
Hydrophobicity	0.902	0.920	0.884	0.804	0.943
Isoelectric point	0.887	0.931	0.843	0.777	0.941
Molecular weight	0.863	**0.936**	0.789	0.733	0.948
Zscale+Hydrophobicity+Isoelectric point+Molecular weight (PCP)	0.911	0.919	0.903	0.822	0.951
PCP+BLOSUM45	0.923	0.902	**0.943**	0.846	0.953
PCP+BLOSUM80	0.923	0.908	0.937	0.845	0.954
PCP+PAM30	0.921	0.909	0.932	0.842	0.954
PCP+PAM70	0.923	0.909	0.936	0.845	0.950
PCP+PAM250	0.921	0.907	0.936	0.843	0.951
PCP+BLOSUM62	0.926	0.909	0.942	0.852	0.958

Table S6. Ablation analysis of model architecture components.

Ablation	Accuracy	Sensitivity	Specificity	MCC	AUC
w/o Sequence Features	0.883	0.811	0.954	0.773	0.948
w/o Graph Transformer	0.906	0.918	0.895	0.813	0.948
w/o Graph Attention	0.912	0.913	0.910	0.823	0.952
w/o Contrastive Learning	0.915	0.912	0.919	0.831	0.950
w/o Seq-Struct Contrast	0.918	0.907	0.930	0.837	0.954
w/o Label Contrast	0.922	0.908	0.936	0.844	0.954
GeoCTP (Full model)	0.926	0.909	0.942	0.852	0.958

Table S7. Comparison between single-task and multi-task learning frameworks on binary classification. The best performance in each metric is highlighted in bold.

Method	Accuracy	Sensitivity	Specificity	MCC	AUC
Single-task	**0.926**	**0.909**	**0.942**	**0.852**	**0.958**
Multi-task	0.907	0.908	0.907	0.815	0.950

Table S8. Comparison between single-task and multi-task learning frameworks on multi-label classification. The best performance in each metric is highlighted in bold.

Method	Aiming	Coverage	$\text{Accuracy}_{\text{MLC}}$	Absolute True	Absolute False
Single-task	**0.905**	**0.858**	**0.829**	**0.717**	**0.075**
Multi-task	0.870	0.846	0.801	0.689	0.091

Table S9. Evaluation of different negative sampling strategies.

Method	Accuracy	Sensitivity	Specificity	MCC	AUC
Random	0.934	0.918	0.950	0.869	0.964
Similarity	0.926	0.909	0.942	0.852	0.958
Hard-negative	0.913	0.898	0.929	0.827	0.950

Table S10. Performance comparison under different settings of optimization and training hyperparameters. The best performance in each metric is highlighted in bold.

Learning rate	Weight decay	Batch size	Accuracy	Sensitivity	Specificity	MCC	AUC
1e-3	0	32	0.923	0.901	0.944	0.846	0.956
1e-3	0	64	0.922	0.901	0.943	0.845	0.951
1e-3	0	128	0.925	0.902	0.948	0.851	0.955
1e-3	1e-5	32	0.923	0.898	0.948	0.847	0.958
1e-3	1e-5	64	0.924	0.901	0.948	0.850	0.957

1e-3	1e-5	128	0.922	0.902	0.942	0.844	0.956
1e-3	1e-4	32	0.923	0.904	0.941	0.846	0.956
1e-3	1e-4	64	0.922	0.904	0.939	0.844	0.956
1e-3	1e-4	128	0.922	0.900	0.944	0.845	0.958
5e-4	0	32	0.922	0.900	0.944	0.845	0.958
5e-4	0	64	0.923	0.903	0.942	0.846	0.955
5e-4	0	128	0.924	0.907	0.942	0.849	0.953
5e-4	1e-5	32	**0.926**	0.909	0.942	**0.852**	**0.958**
5e-4	1e-5	64	0.921	0.908	0.933	0.842	0.954
5e-4	1e-5	128	0.922	0.904	0.939	0.844	0.956
5e-4	1e-4	32	0.922	0.904	0.939	0.844	0.958
5e-4	1e-4	64	0.921	0.901	0.942	0.843	0.954
5e-4	1e-4	128	0.924	0.903	0.944	0.848	0.952
3e-4	0	32	0.923	0.902	0.944	0.847	0.954
3e-4	0	64	0.921	0.900	0.943	0.843	0.957
3e-4	0	128	0.922	0.901	0.943	0.845	0.955
3e-4	1e-5	32	0.923	**0.910**	0.935	0.845	0.956
3e-4	1e-5	64	0.922	0.903	0.941	0.844	0.955
3e-4	1e-5	128	0.923	0.906	0.939	0.846	0.95
3e-4	1e-4	32	0.924	0.907	0.941	0.848	0.955
3e-4	1e-4	64	0.922	0.906	0.938	0.844	0.956
3e-4	1e-4	128	0.923	0.896	**0.949**	0.846	0.955
1e-4	0	32	0.924	0.901	0.948	0.85	0.955
1e-4	0	64	0.924	0.907	0.942	0.849	0.953
1e-4	0	128	0.923	0.906	0.939	0.846	0.95
1e-4	1e-5	32	0.923	0.904	0.941	0.846	0.954
1e-4	1e-5	64	0.924	0.909	0.938	0.848	0.956

1e-4	1e-5	128	0.923	0.908	0.938	0.847	0.952
1e-4	1e-4	32	0.923	0.909	0.937	0.847	0.955
1e-4	1e-4	64	0.924	0.901	0.947	0.848	0.951
1e-4	1e-4	128	0.924	0.908	0.941	0.849	0.947
1e-3	0	32	0.923	0.901	0.944	0.846	0.956
1e-3	0	64	0.922	0.901	0.943	0.845	0.951
1e-3	0	128	0.925	0.902	0.948	0.851	0.955
1e-3	1e-5	32	0.923	0.898	0.948	0.847	0.958
1e-3	1e-5	64	0.924	0.901	0.948	0.85	0.957
1e-3	1e-5	128	0.922	0.902	0.942	0.844	0.956
1e-3	1e-4	32	0.923	0.904	0.941	0.846	0.956
1e-3	1e-4	64	0.922	0.904	0.939	0.844	0.956
1e-3	1e-4	128	0.922	0.900	0.944	0.845	0.958
5e-4	0	32	0.922	0.900	0.944	0.845	0.958
5e-4	0	64	0.923	0.903	0.942	0.846	0.955
5e-4	0	128	0.924	0.907	0.942	0.849	0.953
5e-4	1e-5	32	0.921	0.903	0.938	0.842	0.954
5e-4	1e-5	64	0.921	0.908	0.933	0.842	0.954
5e-4	1e-5	128	0.926	0.909	0.942	0.852	0.958
5e-4	1e-4	32	0.922	0.904	0.939	0.844	0.958
5e-4	1e-4	64	0.921	0.901	0.942	0.843	0.954
5e-4	1e-4	128	0.924	0.903	0.944	0.848	0.952
3e-4	0	32	0.923	0.902	0.944	0.847	0.954
3e-4	0	64	0.921	0.900	0.943	0.843	0.957
3e-4	0	128	0.922	0.901	0.943	0.845	0.955
3e-4	1e-5	32	0.923	0.910	0.935	0.845	0.956
3e-4	1e-5	64	0.922	0.903	0.941	0.844	0.955

3e-4	1e-5	128	0.923	0.906	0.939	0.846	0.95
3e-4	1e-4	32	0.924	0.907	0.941	0.848	0.955
3e-4	1e-4	64	0.922	0.906	0.938	0.844	0.956
3e-4	1e-4	128	0.923	0.896	0.949	0.846	0.955
1e-4	0	32	0.924	0.901	0.948	0.85	0.955
1e-4	0	64	0.924	0.907	0.942	0.849	0.953
1e-4	0	128	0.923	0.906	0.939	0.846	0.95
1e-4	1e-5	32	0.923	0.904	0.941	0.846	0.954
1e-4	1e-5	64	0.924	0.909	0.938	0.848	0.956
1e-4	1e-5	128	0.923	0.908	0.938	0.847	0.952
1e-4	1e-4	32	0.923	0.909	0.937	0.847	0.955
1e-4	1e-4	64	0.924	0.901	0.947	0.848	0.951
1e-4	1e-4	128	0.924	0.908	0.941	0.849	0.947

Table S11. Performance comparison under different spatial cutoff distances for protein graph construction. The best performance in each metric is highlighted in bold.

Cutoff threshold (Å)	Accuracy	Sensitivity	Specificity	MCC	AUC
6	0.912	**0.915**	0.908	0.823	0.950
8	**0.926**	0.909	**0.942**	**0.852**	**0.958**
10	0.908	**0.915**	0.901	0.816	0.947
12	0.913	0.910	0.915	0.826	0.947

Table S12. Performance comparison of graph transformer architectures with varying layers and attention heads. The best performance in each metric is highlighted in bold.

Number of layers (L)	Number of heads (H)	Accuracy	Sensitivity	Specificity	MCC	AUC
1	4	0.916	0.909	0.923	0.832	0.950
2	4	0.916	0.908	0.924	0.832	0.951
3	4	0.906	0.915	0.896	0.811	0.952
4	4	0.923	0.903	0.942	0.846	0.950
1	8	0.916	0.909	0.923	0.832	0.951
2	8	**0.926**	0.909	**0.942**	**0.852**	**0.958**
3	8	0.919	0.901	0.938	0.840	0.950
4	8	0.904	**0.919**	0.889	0.808	0.951
1	16	0.917	0.913	0.921	0.834	0.951
2	16	0.902	0.915	0.889	0.804	0.948
3	16	0.917	0.913	0.921	0.834	0.951
4	16	0.919	0.910	0.927	0.838	0.949

Table S13. Performance comparison under different settings of contrastive learning hyperparameters. The best performance in each metric is highlighted in bold.

Temperature parameter τ	W_{seq}	W_{lab}	Accuracy	Sensitivity	Specificity	MCC	AUC
0.05	0.1	0.1	0.914	0.913	0.915	0.828	0.949
0.05	0.1	0.2	0.922	0.914	0.930	0.844	0.952
0.05	0.1	0.3	0.918	0.908	0.927	0.836	0.951
0.05	0.1	0.4	0.919	0.910	0.929	0.839	0.953
0.05	0.1	0.5	0.923	0.901	0.946	0.847	0.951
0.05	0.1	0.6	0.904	0.918	0.891	0.809	0.950
0.05	0.1	0.7	0.893	0.926	0.861	0.789	0.948
0.05	0.1	0.8	0.920	0.909	0.931	0.840	0.951
0.05	0.1	0.9	0.912	0.915	0.909	0.824	0.951

0.05	0.1	1	0.916	0.914	0.918	0.832	0.949
0.05	0.2	0.1	0.904	0.920	0.889	0.809	0.952
0.05	0.2	0.2	0.914	0.915	0.913	0.828	0.952
0.05	0.2	0.3	0.901	0.920	0.881	0.802	0.950
0.05	0.2	0.4	0.924	0.901	0.947	0.848	0.952
0.05	0.2	0.5	0.919	0.909	0.930	0.839	0.952
0.05	0.2	0.6	0.915	0.916	0.913	0.829	0.952
0.05	0.2	0.7	0.914	0.914	0.914	0.828	0.952
0.05	0.2	0.8	0.916	0.909	0.924	0.833	0.952
0.05	0.2	0.9	0.886	0.926	0.846	0.775	0.951
0.05	0.2	1	0.907	0.918	0.896	0.814	0.948
0.05	0.3	0.1	0.882	0.926	0.838	0.767	0.948
0.05	0.3	0.2	0.923	0.902	0.944	0.847	0.953
0.05	0.3	0.3	0.923	0.900	0.946	0.846	0.951
0.05	0.3	0.4	0.923	0.908	0.937	0.845	0.948
0.05	0.3	0.5	0.923	0.903	0.943	0.847	0.951
0.05	0.3	0.6	0.911	0.918	0.904	0.822	0.951
0.05	0.3	0.7	0.917	0.913	0.921	0.834	0.952
0.05	0.3	0.8	0.918	0.912	0.924	0.835	0.951
0.05	0.3	0.9	0.925	0.896	0.954	0.851	0.952
0.05	0.3	1	0.914	0.909	0.919	0.828	0.95
0.05	0.4	0.1	0.892	0.929	0.855	0.785	0.95
0.05	0.4	0.2	0.915	0.910	0.920	0.831	0.952
0.05	0.4	0.3	0.883	0.921	0.845	0.769	0.949
0.05	0.4	0.4	0.915	0.909	0.921	0.831	0.951
0.05	0.4	0.5	0.919	0.907	0.931	0.838	0.951
0.05	0.4	0.6	0.914	0.914	0.914	0.828	0.952

0.05	0.4	0.7	0.922	0.904	0.939	0.844	0.952
0.05	0.4	0.8	0.919	0.906	0.933	0.839	0.954
0.05	0.4	0.9	0.919	0.907	0.932	0.839	0.953
0.05	0.4	1	0.918	0.909	0.927	0.837	0.951
0.05	0.5	0.1	0.901	0.924	0.878	0.802	0.948
0.05	0.5	0.2	0.923	0.908	0.938	0.847	0.951
0.05	0.5	0.3	0.919	0.909	0.930	0.839	0.948
0.05	0.5	0.4	0.922	0.908	0.936	0.844	0.952
0.05	0.5	0.5	0.896	0.919	0.873	0.793	0.949
0.05	0.5	0.6	0.919	0.910	0.929	0.839	0.950
0.05	0.5	0.7	0.915	0.912	0.918	0.829	0.952
0.05	0.5	0.8	0.918	0.913	0.923	0.835	0.951
0.05	0.5	0.9	0.915	0.914	0.915	0.829	0.953
0.05	0.5	1	0.916	0.909	0.923	0.832	0.951
0.05	0.6	0.1	0.910	0.919	0.902	0.821	0.951
0.05	0.6	0.2	0.906	0.918	0.895	0.813	0.951
0.05	0.6	0.3	0.919	0.913	0.925	0.838	0.950
0.05	0.6	0.4	0.925	0.896	0.954	0.851	0.950
0.05	0.6	0.5	0.919	0.901	0.937	0.838	0.952
0.05	0.6	0.6	0.919	0.902	0.937	0.84	0.95
0.05	0.6	0.7	0.924	0.907	0.942	0.849	0.952
0.05	0.6	0.8	0.913	0.902	0.925	0.827	0.95
0.05	0.6	0.9	0.908	0.916	0.9	0.816	0.951
0.05	0.6	1	0.924	0.9	0.948	0.848	0.951
0.05	0.7	0.1	0.914	0.912	0.916	0.828	0.952
0.05	0.7	0.2	0.919	0.91	0.927	0.838	0.949
0.05	0.7	0.3	0.915	0.91	0.92	0.831	0.954

0.05	0.7	0.4	0.921	0.912	0.931	0.843	0.95
0.05	0.7	0.5	0.920	0.906	0.935	0.841	0.951
0.05	0.7	0.6	0.921	0.909	0.932	0.842	0.95
0.05	0.7	0.7	0.920	0.908	0.932	0.84	0.948
0.05	0.7	0.8	0.916	0.91	0.921	0.832	0.95
0.05	0.7	0.9	0.918	0.91	0.926	0.837	0.948
0.05	0.7	1	0.918	0.907	0.93	0.837	0.949
0.05	0.8	0.1	0.902	0.916	0.887	0.804	0.951
0.05	0.8	0.2	0.924	0.902	0.946	0.848	0.952
0.05	0.8	0.3	0.910	0.92	0.901	0.821	0.949
0.05	0.8	0.4	0.923	0.907	0.939	0.847	0.951
0.05	0.8	0.5	0.918	0.913	0.924	0.837	0.952
0.05	0.8	0.6	0.920	0.913	0.927	0.84	0.951
0.05	0.8	0.7	0.921	0.895	0.948	0.844	0.949
0.05	0.8	0.8	0.919	0.913	0.926	0.839	0.951
0.05	0.8	0.9	0.924	0.902	0.946	0.848	0.952
0.05	0.8	1	0.923	0.907	0.938	0.845	0.953
0.05	0.9	0.1	0.919	0.904	0.933	0.838	0.948
0.05	0.9	0.2	0.916	0.909	0.924	0.833	0.952
0.05	0.9	0.3	0.922	0.896	0.948	0.845	0.948
0.05	0.9	0.4	0.889	0.921	0.856	0.779	0.952
0.05	0.9	0.5	0.921	0.908	0.933	0.842	0.949
0.05	0.9	0.6	0.923	0.906	0.941	0.847	0.953
0.05	0.9	0.7	0.915	0.914	0.915	0.829	0.950
0.05	0.9	0.8	0.912	0.918	0.907	0.825	0.950
0.05	0.9	0.9	0.893	0.925	0.862	0.788	0.951
0.05	0.9	1	0.921	0.907	0.936	0.843	0.948

0.05	1	0.1	0.919	0.902	0.937	0.840	0.950
0.05	1	0.2	0.910	0.919	0.901	0.82	0.951
0.05	1	0.3	0.879	0.929	0.829	0.762	0.952
0.05	1	0.4	0.910	0.919	0.901	0.820	0.950
0.05	1	0.5	0.918	0.915	0.92	0.835	0.949
0.05	1	0.6	0.915	0.914	0.915	0.829	0.952
0.05	1	0.7	0.912	0.915	0.908	0.823	0.948
0.05	1	0.8	0.915	0.909	0.920	0.829	0.953
0.05	1	0.9	0.916	0.912	0.920	0.832	0.950
0.05	1	1	0.915	0.913	0.918	0.831	0.95
0.1	0.1	0.1	0.895	0.927	0.863	0.792	0.946
0.1	0.1	0.2	0.919	0.909	0.929	0.838	0.951
0.1	0.1	0.3	0.909	0.914	0.904	0.818	0.949
0.1	0.1	0.4	0.920	0.910	0.930	0.840	0.951
0.1	0.1	0.5	0.924	0.904	0.943	0.848	0.952
0.1	0.1	0.6	0.913	0.920	0.906	0.826	0.950
0.1	0.1	0.7	0.915	0.910	0.920	0.831	0.951
0.1	0.1	0.8	0.910	0.918	0.902	0.820	0.953
0.1	0.1	0.9	0.922	0.902	0.942	0.844	0.953
0.1	0.1	1	0.915	0.907	0.924	0.831	0.949
0.1	0.2	0.1	0.915	0.910	0.920	0.831	0.953
0.1	0.2	0.2	0.913	0.914	0.912	0.826	0.947
0.1	0.2	0.3	0.920	0.907	0.933	0.840	0.951
0.1	0.2	0.4	0.917	0.915	0.919	0.834	0.952
0.1	0.2	0.5	0.920	0.910	0.930	0.840	0.949
0.1	0.2	0.6	0.919	0.904	0.935	0.839	0.948
0.1	0.2	0.7	0.919	0.909	0.929	0.838	0.952

0.1	0.2	0.8	0.924	0.887	**0.960**	0.850	0.951
0.1	0.2	0.9	0.920	0.898	0.942	0.841	0.953
0.1	0.2	1	0.915	0.908	0.921	0.829	0.953
0.1	0.3	0.1	0.918	0.910	0.925	0.835	0.951
0.1	0.3	0.2	0.923	0.904	0.941	0.846	0.948
0.1	0.3	0.3	0.915	0.912	0.918	0.829	0.953
0.1	0.3	0.4	0.892	0.918	0.867	0.786	0.948
0.1	0.3	0.5	0.924	0.903	0.946	0.849	0.951
0.1	0.3	0.6	0.923	0.901	0.944	0.846	0.952
0.1	0.3	0.7	0.910	0.920	0.900	0.820	0.949
0.1	0.3	0.8	0.923	0.901	0.944	0.846	0.954
0.1	0.3	0.9	0.913	0.915	0.912	0.827	0.952
0.1	0.3	1	0.910	0.915	0.904	0.82	0.949
0.1	0.4	0.1	0.915	0.91	0.919	0.829	0.952
0.1	0.4	0.2	0.912	0.913	0.912	0.824	0.951
0.1	0.4	0.3	0.916	0.914	0.919	0.833	0.949
0.1	0.4	0.4	0.912	0.916	0.907	0.823	0.946
0.1	0.4	0.5	0.905	0.915	0.895	0.810	0.949
0.1	0.4	0.6	0.921	0.909	0.932	0.842	0.950
0.1	0.4	0.7	0.867	0.933	0.800	0.740	0.948
0.1	0.4	0.8	0.895	0.921	0.868	0.790	0.946
0.1	0.4	0.9	0.908	0.916	0.900	0.816	0.950
0.1	0.4	1	0.921	0.907	0.935	0.842	0.954
0.1	0.5	0.1	0.898	0.923	0.874	0.798	0.952
0.1	0.5	0.2	0.912	0.920	0.904	0.825	0.948
0.1	0.5	0.3	0.918	0.912	0.924	0.835	0.951
0.1	0.5	0.4	0.918	0.912	0.925	0.837	0.950

0.1	0.5	0.5	0.923	0.900	0.946	0.846	0.951
0.1	0.5	0.6	0.913	0.918	0.908	0.826	0.951
0.1	0.5	0.7	0.921	0.903	0.938	0.842	0.951
0.1	0.5	0.8	0.910	0.914	0.907	0.821	0.952
0.1	0.5	0.9	0.904	0.919	0.890	0.809	0.949
0.1	0.5	1	0.921	0.909	0.933	0.843	0.950
0.1	0.6	0.1	0.918	0.908	0.927	0.836	0.953
0.1	0.6	0.2	0.910	0.914	0.907	0.821	0.952
0.1	0.6	0.3	0.919	0.906	0.933	0.839	0.951
0.1	0.6	0.4	0.915	0.916	0.913	0.829	0.952
0.1	0.6	0.5	0.910	0.921	0.900	0.821	0.946
0.1	0.6	0.6	0.919	0.902	0.937	0.84	0.952
0.1	0.6	0.7	0.856	**0.938**	0.774	0.722	0.951
0.1	0.6	0.8	0.904	0.921	0.887	0.809	0.950
0.1	0.6	0.9	0.922	0.906	0.938	0.844	0.951
0.1	0.6	1	0.919	0.908	0.931	0.839	0.948
0.1	0.7	0.1	0.921	0.906	0.936	0.842	0.951
0.1	0.7	0.2	0.916	0.909	0.923	0.832	0.951
0.1	0.7	0.3	0.921	0.898	0.944	0.844	0.952
0.1	0.7	0.4	0.912	0.914	0.910	0.824	0.948
0.1	0.7	0.5	0.922	0.902	0.942	0.844	0.952
0.1	0.7	0.6	0.904	0.920	0.887	0.808	0.944
0.1	0.7	0.7	0.919	0.900	0.939	0.840	0.947
0.1	0.7	0.8	0.906	0.915	0.897	0.812	0.951
0.1	0.7	0.9	0.895	0.920	0.869	0.790	0.950
0.1	0.7	1	0.906	0.918	0.893	0.811	0.949
0.1	0.8	0.1	0.911	0.908	0.914	0.822	0.951

0.1	0.8	0.2	0.916	0.915	0.918	0.833	0.949
0.1	0.8	0.3	0.905	0.918	0.892	0.810	0.953
0.1	0.8	0.4	0.909	0.913	0.906	0.818	0.950
0.1	0.8	0.5	0.914	0.910	0.918	0.828	0.953
0.1	0.8	0.6	0.917	0.913	0.921	0.834	0.951
0.1	0.8	0.7	0.893	0.924	0.863	0.788	0.949
0.1	0.8	0.8	0.915	0.91	0.919	0.829	0.951
0.1	0.8	0.9	0.908	0.913	0.903	0.816	0.953
0.1	0.8	1	0.915	0.910	0.919	0.829	0.951
0.1	0.9	0.1	0.920	0.907	0.933	0.840	0.949
0.1	0.9	0.2	0.921	0.904	0.937	0.842	0.952
0.1	0.9	0.3	0.910	0.914	0.906	0.82	0.947
0.1	0.9	0.4	0.918	0.909	0.926	0.835	0.951
0.1	0.9	0.5	0.918	0.909	0.927	0.837	0.948
0.1	0.9	0.6	0.902	0.918	0.886	0.804	0.950
0.1	0.9	0.7	0.919	0.908	0.930	0.838	0.952
0.1	0.9	0.8	0.894	0.924	0.864	0.790	0.946
0.1	0.9	0.9	0.916	0.906	0.927	0.833	0.947
0.1	0.9	1	0.916	0.910	0.921	0.832	0.953
0.1	1	0.1	0.918	0.902	0.935	0.837	0.95
0.1	1	0.2	0.903	0.920	0.885	0.806	0.947
0.1	1	0.3	0.920	0.910	0.930	0.840	0.950
0.1	1	0.4	0.916	0.909	0.923	0.832	0.954
0.1	1	0.5	0.919	0.910	0.929	0.839	0.952
0.1	1	0.6	0.904	0.919	0.890	0.809	0.95
0.1	1	0.7	0.925	0.901	0.949	0.851	0.952
0.1	1	0.8	0.923	0.908	0.938	0.847	0.952

0.1	1	0.9	0.895	0.919	0.872	0.791	0.951
0.1	1	1	0.894	0.923	0.866	0.789	0.949
0.15	0.1	0.1	0.913	0.915	0.912	0.827	0.953
0.15	0.1	0.2	0.903	0.919	0.886	0.806	0.948
0.15	0.1	0.3	0.923	0.898	0.947	0.846	0.952
0.15	0.1	0.4	0.921	0.906	0.936	0.842	0.952
0.15	0.1	0.5	0.910	0.918	0.902	0.820	0.948
0.15	0.1	0.6	0.924	0.896	0.953	0.850	0.953
0.15	0.1	0.7	0.915	0.914	0.915	0.829	0.950
0.15	0.1	0.8	0.921	0.902	0.939	0.842	0.949
0.15	0.1	0.9	0.919	0.908	0.930	0.838	0.950
0.15	0.1	1	0.918	0.909	0.927	0.837	0.952
0.15	0.2	0.1	0.920	0.909	0.931	0.840	0.953
0.15	0.2	0.2	0.923	0.908	0.938	0.847	0.952
0.15	0.2	0.3	0.856	0.935	0.777	0.721	0.947
0.15	0.2	0.4	0.918	0.909	0.926	0.835	0.953
0.15	0.2	0.5	0.920	0.908	0.932	0.840	0.952
0.15	0.2	0.6	0.920	0.908	0.932	0.840	0.953
0.15	0.2	0.7	0.923	0.903	0.942	0.846	0.950
0.15	0.2	0.8	0.923	0.906	0.939	0.846	0.951
0.15	0.2	0.9	0.901	0.916	0.885	0.802	0.942
0.15	0.2	1	0.919	0.908	0.93	0.838	0.950
0.15	0.3	0.1	0.916	0.908	0.924	0.832	0.952
0.15	0.3	0.2	0.894	0.924	0.864	0.790	0.946
0.15	0.3	0.3	0.923	0.901	0.944	0.846	0.949
0.15	0.3	0.4	0.921	0.909	0.933	0.843	0.951
0.15	0.3	0.5	0.917	0.912	0.923	0.834	0.951

0.15	0.3	0.6	0.910	0.918	0.902	0.820	0.951
0.15	0.3	0.7	0.918	0.908	0.927	0.836	0.951
0.15	0.3	0.8	0.918	0.909	0.926	0.835	0.949
0.15	0.3	0.9	0.923	0.903	0.942	0.846	0.949
0.15	0.3	1	0.921	0.895	0.948	0.844	0.953
0.15	0.4	0.1	0.921	0.907	0.936	0.843	0.947
0.15	0.4	0.2	0.917	0.909	0.925	0.834	0.952
0.15	0.4	0.3	0.905	0.915	0.895	0.81	0.951
0.15	0.4	0.4	0.922	0.909	0.935	0.844	0.950
0.15	0.4	0.5	0.909	0.915	0.903	0.818	0.952
0.15	0.4	0.6	0.916	0.910	0.921	0.832	0.951
0.15	0.4	0.7	0.893	0.919	0.867	0.787	0.952
0.15	0.4	0.8	0.925	0.898	0.952	0.851	0.951
0.15	0.4	0.9	0.916	0.919	0.913	0.832	0.953
0.15	0.4	1	0.921	0.908	0.933	0.842	0.948
0.15	0.5	0.1	0.918	0.910	0.925	0.835	0.951
0.15	0.5	0.2	0.921	0.906	0.937	0.843	0.951
0.15	0.5	0.3	0.92	0.896	0.944	0.841	0.954
0.15	0.5	0.4	0.921	0.908	0.935	0.843	0.950
0.15	0.5	0.5	0.907	0.921	0.892	0.814	0.950
0.15	0.5	0.6	0.914	0.912	0.916	0.828	0.950
0.15	0.5	0.7	0.922	0.897	0.947	0.845	0.952
0.15	0.5	0.8	0.904	0.918	0.891	0.809	0.953
0.15	0.5	0.9	0.919	0.903	0.935	0.838	0.952
0.15	0.5	1	0.919	0.910	0.927	0.838	0.953
0.15	0.6	0.1	0.919	0.909	0.929	0.838	0.949
0.15	0.6	0.2	0.923	0.903	0.943	0.847	0.951

0.15	0.6	0.3	0.913	0.919	0.907	0.826	0.95
0.15	0.6	0.4	0.923	0.903	0.942	0.846	0.951
0.15	0.6	0.5	0.912	0.916	0.908	0.824	0.950
0.15	0.6	0.6	0.912	0.913	0.91	0.823	0.948
0.15	0.6	0.7	0.918	0.906	0.931	0.837	0.952
0.15	0.6	0.8	0.918	0.908	0.929	0.837	0.947
0.15	0.6	0.9	0.909	0.908	0.909	0.817	0.944
0.15	0.6	1	0.925	0.897	0.953	0.851	0.951
0.15	0.7	0.1	0.919	0.906	0.933	0.839	0.949
0.15	0.7	0.2	0.921	0.909	0.933	0.843	0.953
0.15	0.7	0.3	0.913	0.918	0.908	0.826	0.950
0.15	0.7	0.4	0.921	0.907	0.936	0.843	0.951
0.15	0.7	0.5	0.916	0.912	0.920	0.832	0.951
0.15	0.7	0.6	0.918	0.913	0.923	0.835	0.948
0.15	0.7	0.7	0.924	0.908	0.939	0.848	0.952
0.15	0.7	0.8	0.909	0.918	0.901	0.819	0.95
0.15	0.7	0.9	0.915	0.914	0.916	0.831	0.951
0.15	0.7	1	0.917	0.914	0.920	0.834	0.951
0.15	0.8	0.1	0.889	0.923	0.856	0.780	0.950
0.15	0.8	0.2	0.922	0.906	0.938	0.844	0.953
0.15	0.8	0.3	0.904	0.919	0.889	0.808	0.952
0.15	0.8	0.4	0.904	0.921	0.886	0.808	0.947
0.15	0.8	0.5	0.921	0.910	0.932	0.843	0.950
0.15	0.8	0.6	0.916	0.914	0.919	0.833	0.948
0.15	0.8	0.7	0.912	0.919	0.906	0.825	0.949
0.15	0.8	0.8	0.921	0.904	0.938	0.843	0.954
0.15	0.8	0.9	0.915	0.909	0.921	0.831	0.951

0.15	0.8	1	0.919	0.908	0.931	0.839	0.950
0.15	0.9	0.1	0.916	0.910	0.921	0.832	0.947
0.15	0.9	0.2	0.92	0.900	0.941	0.841	0.951
0.15	0.9	0.3	0.916	0.909	0.923	0.832	0.951
0.15	0.9	0.4	0.896	0.923	0.870	0.794	0.949
0.15	0.9	0.5	0.904	0.919	0.890	0.809	0.947
0.15	0.9	0.6	0.911	0.916	0.906	0.822	0.951
0.15	0.9	0.7	0.918	0.915	0.920	0.835	0.947
0.15	0.9	0.8	0.920	0.907	0.933	0.840	0.949
0.15	0.9	0.9	0.907	0.918	0.896	0.814	0.950
0.15	0.9	1	0.892	0.919	0.864	0.784	0.953
0.15	1	0.1	0.919	0.909	0.930	0.839	0.950
0.15	1	0.2	0.916	0.912	0.921	0.833	0.947
0.15	1	0.3	0.921	0.902	0.941	0.843	0.950
0.15	1	0.4	0.920	0.909	0.931	0.84	0.953
0.15	1	0.5	0.915	0.909	0.921	0.831	0.952
0.15	1	0.6	0.916	0.910	0.921	0.832	0.953
0.15	1	0.7	0.923	0.907	0.938	0.845	0.948
0.15	1	0.8	0.922	0.906	0.938	0.844	0.953
0.15	1	0.9	0.919	0.910	0.927	0.838	0.955
0.15	1	1	0.892	0.924	0.861	0.786	0.948
0.2	0.1	0.1	0.909	0.923	0.895	0.818	0.951
0.2	0.1	0.2	0.916	0.908	0.925	0.833	0.951
0.2	0.1	0.3	0.923	0.902	0.943	0.846	0.949
0.2	0.1	0.4	0.913	0.912	0.915	0.827	0.953
0.2	0.1	0.5	0.914	0.921	0.907	0.828	0.950
0.2	0.1	0.6	0.918	0.903	0.933	0.837	0.952

0.2	0.1	0.7	0.897	0.920	0.874	0.795	0.950
0.2	0.1	0.8	0.918	0.902	0.933	0.836	0.947
0.2	0.1	0.9	0.919	0.910	0.929	0.839	0.951
0.2	0.1	1	0.913	0.912	0.915	0.827	0.952
0.2	0.2	0.1	0.921	0.902	0.941	0.843	0.952
0.2	0.2	0.2	0.922	0.908	0.936	0.844	0.951
0.2	0.2	0.3	0.924	0.907	0.942	0.849	0.951
0.2	0.2	0.4	0.916	0.913	0.920	0.833	0.950
0.2	0.2	0.5	0.926	0.898	0.953	0.852	0.952
0.2	0.2	0.6	0.924	0.902	0.946	0.848	0.951
0.2	0.2	0.7	0.916	0.915	0.918	0.833	0.952
0.2	0.2	0.8	0.914	0.913	0.915	0.828	0.947
0.2	0.2	0.9	0.872	0.937	0.808	0.751	0.949
0.2	0.2	1	0.917	0.912	0.923	0.834	0.949
0.2	0.3	0.1	0.922	0.908	0.936	0.844	0.954
0.2	0.3	0.2	0.887	0.921	0.854	0.777	0.95
0.2	0.3	0.3	0.922	0.909	0.935	0.844	0.948
0.2	0.3	0.4	0.917	0.907	0.927	0.834	0.952
0.2	0.3	0.5	0.917	0.915	0.919	0.834	0.949
0.2	0.3	0.6	0.913	0.909	0.918	0.827	0.951
0.2	0.3	0.7	0.907	0.92	0.893	0.814	0.949
0.2	0.3	0.8	0.919	0.906	0.932	0.838	0.950
0.2	0.3	0.9	0.912	0.912	0.912	0.823	0.948
0.2	0.3	1	0.912	0.915	0.908	0.823	0.943
0.2	0.4	0.1	0.923	0.903	0.942	0.846	0.951
0.2	0.4	0.2	0.913	0.914	0.912	0.826	0.949
0.2	0.4	0.3	0.916	0.91	0.921	0.832	0.949

0.2	0.4	0.4	0.921	0.912	0.930	0.842	0.950
0.2	0.4	0.5	0.914	0.916	0.912	0.828	0.948
0.2	0.4	0.6	0.912	0.913	0.910	0.823	0.95
0.2	0.4	0.7	0.922	0.906	0.938	0.844	0.954
0.2	0.4	0.8	0.892	0.921	0.862	0.785	0.947
0.2	0.4	0.9	0.913	0.918	0.909	0.827	0.947
0.2	0.4	1	0.917	0.910	0.924	0.834	0.951
0.2	0.5	0.1	0.922	0.908	0.936	0.844	0.952
0.2	0.5	0.2	0.913	0.914	0.913	0.827	0.950
0.2	0.5	0.3	0.880	0.933	0.826	0.764	0.950
0.2	0.5	0.4	0.915	0.921	0.908	0.829	0.948
0.2	0.5	0.5	0.912	0.912	0.913	0.824	0.950
0.2	0.5	0.6	0.921	0.909	0.932	0.842	0.953
0.2	0.5	0.7	0.884	0.929	0.839	0.771	0.947
0.2	0.5	0.8	0.918	0.916	0.920	0.837	0.950
0.2	0.5	0.9	0.921	0.909	0.932	0.842	0.949
0.2	0.5	1	0.911	0.915	0.907	0.822	0.951
0.2	0.6	0.1	0.921	0.902	0.941	0.843	0.952
0.2	0.6	0.2	0.880	0.933	0.826	0.764	0.95
0.2	0.6	0.3	0.915	0.913	0.918	0.831	0.952
0.2	0.6	0.4	**0.926**	0.909	0.942	**0.852**	**0.958**
0.2	0.6	0.5	0.912	0.910	0.914	0.824	0.951
0.2	0.6	0.6	0.907	0.916	0.897	0.814	0.953
0.2	0.6	0.7	0.918	0.908	0.929	0.837	0.951
0.2	0.6	0.8	0.909	0.909	0.909	0.818	0.952
0.2	0.6	0.9	0.921	0.904	0.938	0.843	0.95
0.2	0.6	1	0.922	0.908	0.936	0.844	0.954

0.2	0.7	0.1	0.920	0.909	0.931	0.840	0.949
0.2	0.7	0.2	0.919	0.906	0.932	0.838	0.949
0.2	0.7	0.3	0.922	0.903	0.941	0.844	0.950
0.2	0.7	0.4	0.914	0.914	0.914	0.828	0.953
0.2	0.7	0.5	0.915	0.920	0.910	0.831	0.949
0.2	0.7	0.6	0.918	0.912	0.924	0.835	0.951
0.2	0.7	0.7	0.886	0.926	0.845	0.774	0.951
0.2	0.7	0.8	0.92	0.906	0.935	0.841	0.951
0.2	0.7	0.9	0.919	0.91	0.929	0.839	0.953
0.2	0.7	1	0.911	0.918	0.904	0.822	0.947
0.2	0.8	0.1	0.921	0.907	0.935	0.842	0.952
0.2	0.8	0.2	0.921	0.908	0.935	0.843	0.950
0.2	0.8	0.3	0.916	0.915	0.916	0.832	0.953
0.2	0.8	0.4	0.916	0.912	0.920	0.832	0.950
0.2	0.8	0.5	0.919	0.907	0.931	0.838	0.952
0.2	0.8	0.6	0.923	0.895	0.952	0.848	0.952
0.2	0.8	0.7	0.904	0.916	0.891	0.808	0.951
0.2	0.8	0.8	0.918	0.907	0.930	0.837	0.947
0.2	0.8	0.9	0.923	0.903	0.943	0.847	0.950
0.2	0.8	1	0.919	0.909	0.930	0.839	0.950
0.2	0.9	0.1	0.909	0.915	0.902	0.817	0.951
0.2	0.9	0.2	0.917	0.913	0.921	0.834	0.950
0.2	0.9	0.3	0.912	0.914	0.909	0.823	0.954
0.2	0.9	0.4	0.917	0.91	0.924	0.834	0.950
0.2	0.9	0.5	0.901	0.924	0.878	0.802	0.952
0.2	0.9	0.6	0.924	0.901	0.948	0.85	0.952
0.2	0.9	0.7	0.919	0.909	0.930	0.839	0.951

0.2	0.9	0.8	0.912	0.916	0.907	0.823	0.948
0.2	0.9	0.9	0.916	0.912	0.921	0.833	0.952
0.2	0.9	1	0.921	0.909	0.933	0.843	0.951
0.2	1	0.1	0.892	0.918	0.867	0.786	0.948
0.2	1	0.2	0.904	0.916	0.892	0.809	0.952
0.2	1	0.3	0.88	0.929	0.831	0.763	0.948
0.2	1	0.4	0.912	0.914	0.91	0.824	0.951
0.2	1	0.5	0.924	0.906	0.943	0.849	0.953
0.2	1	0.6	0.919	0.909	0.93	0.839	0.951
0.2	1	0.7	0.916	0.913	0.92	0.833	0.95
0.2	1	0.8	0.922	0.907	0.937	0.844	0.954
0.2	1	0.9	0.916	0.91	0.923	0.833	0.951
0.2	1	1	0.921	0.904	0.938	0.843	0.949

Table S14. Performance comparison of GeoCTP with machine learning baseline models.

Model	Accuracy	Sensitivity	Specificity	MCC	AUC
RandomForest	0.867	0.838	0.896	0.735	0.928
ExtraTrees	0.847	0.851	0.844	0.695	0.915
GradientBoosting	0.886	0.881	0.891	0.772	0.942
AdaBoost	0.850	0.850	0.851	0.701	0.921
KNeighbors	0.729	0.959	0.499	0.515	0.829
GeoCTP	0.926	0.909	0.942	0.852	0.958

Table S15. Performance comparison of GeoCTP with deep learning baseline models.

Model	Accuracy	Sensitivity	Specificity	MCC	AUC
CNN	0.878	0.887	0.869	0.757	0.941
MLP	0.877	0.879	0.874	0.753	0.932
BiLSTM	0.632	0.824	0.439	0.286	0.713
Transformer	0.879	0.832	0.926	0.761	0.932
GCN	0.844	0.841	0.847	0.689	0.903
GAT	0.840	0.833	0.846	0.679	0.901
GeoCTP	0.926	0.909	0.942	0.852	0.958

Table S16. Performance comparison with existing methods on the CTP dataset under the CD-HIT threshold of 0.9, including 95% confidence intervals (CIs) and statistical significance tests.

Method	Accuracy	Sensitivity	Specificity	MCC	Paired t-test P	Wilcoxon P
PEPred-Suite	0.857 (0.837, 0.877)	0.825 (0.794, 0.856)	0.889 (0.863, 0.914)	0.715 (0.676, 0.756)	<0.001	<0.001
PepFormer	0.860 (0.840, 0.879)	0.839 (0.808, 0.868)	0.881 (0.852, 0.906)	0.720 (0.679, 0.759)	<0.001	<0.001
ACP-MHCNN	0.751 (0.727, 0.776)	0.643 (0.602, 0.684)	0.859 (0.828, 0.887)	0.515 (0.466, 0.562)	<0.001	<0.001
GRDF	0.840 (0.820, 0.861)	0.811 (0.779, 0.844)	0.870 (0.844, 0.897)	0.682 (0.642, 0.723)	<0.001	<0.001
ACP-CapsPred	0.860 (0.840, 0.879)	0.835 (0.804, 0.863)	0.885 (0.859, 0.909)	0.721 (0.681, 0.758)	<0.001	<0.001
AntiCP2.0	0.841 (0.820, 0.863)	0.794 (0.760, 0.828)	0.889 (0.863, 0.914)	0.686 (0.644, 0.727)	<0.001	<0.001
AMPfun	0.595 (0.578, 0.613)	0.213 (0.180, 0.245)	0.978 (0.966, 0.990)	0.296 (0.254, 0.335)	<0.001	<0.001
AntiCP3.0	0.871 (0.852, 0.890)	0.837 (0.806, 0.864)	0.906 (0.882, 0.928)	0.744 (0.705, 0.782)	<0.001	<0.001
ACPred-FL	0.825 (0.804, 0.847)	0.774 (0.739, 0.810)	0.877 (0.852, 0.904)	0.654 (0.613, 0.697)	<0.001	<0.001
GeoCTP	0.896 (0.877, 0.913)	0.866 (0.837, 0.894)	0.926 (0.902, 0.947)	0.794 (0.757, 0.828)	-	-

Table S17. Performance comparison with existing methods on the CTP dataset under the CD-HIT threshold of 0.8, including 95% confidence intervals (CIs) and statistical significance tests.

Method	Accuracy	Sensitivity	Specificity	MCC	Paired t-test P	Wilcoxon P
PEPred-Suite	0.824 (0.798, 0.849)	0.784 (0.747, 0.820)	0.864 (0.831, 0.896)	0.651 (0.598, 0.700)	<0.001	<0.001
PepFormer	0.835 (0.810, 0.859)	0.808 (0.769, 0.844)	0.861 (0.827, 0.891)	0.670 (0.621, 0.719)	<0.001	<0.001
ACP-MHCNN	0.740 (0.712, 0.768)	0.582 (0.533, 0.631)	0.898 (0.869, 0.924)	0.506 (0.452, 0.561)	<0.001	<0.001
GRDF	0.826 (0.801, 0.850)	0.804 (0.767, 0.840)	0.847 (0.813, 0.878)	0.652 (0.604, 0.700)	<0.001	<0.001
ACP-CapsPred	0.843 (0.820, 0.866)	0.816 (0.780, 0.849)	0.871 (0.840, 0.900)	0.688 (0.642, 0.732)	<0.001	<0.001
AntiCP2.0	0.816 (0.791, 0.840)	0.780 (0.742, 0.818)	0.851 (0.818, 0.884)	0.633 (0.583, 0.682)	<0.001	<0.001
AMPfun	0.574 (0.557, 0.593)	0.167 (0.131, 0.202)	0.982 (0.969, 0.993)	0.257 (0.207, 0.302)	<0.001	<0.001
AntiCP3.0	0.857 (0.833, 0.879)	0.856 (0.822, 0.887)	0.858 (0.824, 0.889)	0.713 (0.667, 0.758)	<0.001	<0.001
ACPred-FL	0.811 (0.784, 0.837)	0.811 (0.773, 0.849)	0.811 (0.776, 0.847)	0.622 (0.570, 0.674)	<0.001	<0.001
GeoCTP	0.881 (0.860, 0.901)	0.880 (0.849, 0.909)	0.882 (0.851, 0.911)	0.762 (0.720, 0.802)	-	-

Table S18. Paired McNemar tests (GeoCTP vs ACP-CapsPred) across three test sets

CD-HIT Threshold (CTP)	Discordant share (D/N)	ACP-CapsPred-wrong GeoCTP-correct (c)	ACP-CapsPred-correct GeoCTP-wrong (b)	Net gain (c-b)	McNemar *p*
1.0	13.62% (D=225)	136	89	47	2.095×10^{-3}
0.9	18.35% (D=214)	128	86	42	4.950×10^{-3}
0.8	20.00% (D=180)	107	73	34	1.368×10^{-2}

Notes. "Discordant share" denotes the percentage of test samples on which the two models disagree; D is the number of discordant pairs. Exact *p*-values are from the binomial formulation of McNemar's test applied to the discordant counts.

Table S19. Overview of the ACP dataset used as external validation benchmark.

Dataset	Training data			Testing data		
	ACP	Non-ACP	Total	ACP	Non-ACP	Total
Set 1	632	641	1273	161	158	319
Set 2	716	677	1393	186	170	356

Table S20. Comparative evaluation of GeoCTP and state-of-the-art models on ACP set 1, including including 95% confidence intervals (CIs) and statistical significance tests.

Method	Accuracy	Sensitivity	Specificity	MCC	Paired *t*-test *P*	Wilcoxon *P*
PEPred-Suite	0.746 (0.699, 0.790)	0.733 (0.665, 0.795)	0.760 (0.690, 0.823)	0.493 (0.398, 0.581)	<0.001	<0.001
PepFormer	0.715 (0.665, 0.762)	0.640 (0.565, 0.714)	0.791 (0.728, 0.854)	0.436 (0.335, 0.535)	<0.001	<0.001
ACP-MHCNN	0.571 (0.536, 0.605)	0.944 (0.907, 0.975)	0.190 (0.133, 0.253)	0.204 (0.101, 0.296)	<0.001	<0.001
GRDF	0.771 (0.724, 0.818)	0.783 (0.720, 0.845)	0.759 (0.690, 0.823)	0.542 (0.449, 0.636)	<0.001	<0.001
ACP-CapsPred	0.803 (0.759, 0.846)	0.781 (0.720, 0.845)	0.824 (0.766, 0.880)	0.606 (0.518, 0.693)	0.4336	0.4328
AntiCP2.0	0.708 (0.658, 0.759)	0.621 (0.546, 0.696)	0.797 (0.734, 0.854)	0.425 (0.324, 0.526)	<0.001	<0.001
AMPfun	0.687 (0.636, 0.737)	0.720 (0.652, 0.789)	0.652 (0.576, 0.728)	0.373 (0.273, 0.474)	<0.001	<0.001
AntiCP3.0	0.712 (0.661, 0.759)	0.689 (0.615, 0.758)	0.734 (0.665, 0.804)	0.424 (0.323, 0.518)	<0.001	<0.001
ACPred-FL	0.715 (0.665, 0.762)	0.708 (0.640, 0.776)	0.722 (0.652, 0.791)	0.430 (0.331, 0.524)	<0.001	<0.001
GeoCTP	0.815 (0.771, 0.856)	0.876 (0.826, 0.925)	0.753 (0.684, 0.816)	0.634 (0.552, 0.716)	-	-

Table S21. Comparative evaluation of GeoCTP and state-of-the-art models on ACP set 2, including including 95% confidence intervals (CIs) and statistical significance tests.

Method	Accuracy	Sensitivity	Specificity	MCC	Paired *t*-test *P*	Wilcoxon *P*
PEPred-Suite	0.933 (0.904, 0.958)	0.903 (0.855, 0.946)	0.965 (0.935, 0.988)	0.867 (0.815, 0.917)	0.0015	0.0016
PepFormer	0.933 (0.904, 0.958)	0.930 (0.892, 0.962)	0.935 (0.894, 0.971)	0.865 (0.809, 0.916)	0.0015	0.0016
ACP-MHCNN	0.916 (0.885, 0.944)	0.882 (0.833, 0.925)	0.953 (0.924, 0.982)	0.834 (0.776, 0.888)	<0.001	<0.001
GRDF	0.941 (0.916, 0.963)	0.909 (0.866, 0.946)	0.976 (0.953, 0.994)	0.885 (0.837, 0.929)	0.0080	0.0082
ACP-CapsPred	0.955 (0.933, 0.975)	0.952 (0.919, 0.978)	0.959 (0.929, 0.988)	0.910 (0.867, 0.950)	0.3180	0.3173
AntiCP2.0	0.916 (0.888, 0.947)	0.914 (0.871, 0.952)	0.918 (0.876, 0.959)	0.831 (0.775, 0.893)	<0.001	<0.001
AMPfun	0.772 (0.730, 0.812)	0.640 (0.570, 0.710)	0.918 (0.871, 0.959)	0.576 (0.495, 0.649)	<0.001	<0.001
AntiCP3.0	0.949 (0.927, 0.972)	0.930 (0.892, 0.962)	0.971 (0.941, 0.994)	0.900 (0.855, 0.944)	0.0453	0.0455
ACPred-FL	0.916 (0.885, 0.944)	0.882 (0.833, 0.925)	0.953 (0.918, 0.982)	0.834 (0.777, 0.889)	<0.001	<0.001
GeoCTP	0.961 (0.941, 0.980)	0.946 (0.909, 0.973)	0.976 (0.953, 0.994)	0.922 (0.882, 0.961)	-	-